# CARDIO TWIN

## Synthetic patient research report

Research prototype - not a certified medical device. No prospective cardiac-event prediction is provided.

Patient: SYNTH-DEMO-001 | Age: 58 | Gender: Female

Profile and wearable trajectory in this report are invented demonstration data.

## Clinical profile and baseline classification

BMI 29.0; recorded diabetes 0; smoking history never; sleep_hours 6.5. Source coding is unverified.

heart_disease: 0.023. Classification of recorded status, not future event probability.

Input completeness: 82%; imputed fields: nightly_temperature, sleep_points_percentage, TENSE/ANXIOUS, TIRED, ENTERTAINMENT.

Outside training range: none detected.

No validated individual confidence interval. Aggregate bootstrap CIs quantify held-out performance only; range checks cannot prove in-distribution status.

hypertension: 0.115. Classification of recorded status, not future event probability.

Input completeness: 82%; imputed fields: nightly_temperature, sleep_points_percentage, TENSE/ANXIOUS, TIRED, ENTERTAINMENT.

Outside training range: none detected.

No validated individual confidence interval. Aggregate bootstrap CIs quantify held-out performance only; range checks cannot prove in-distribution status.

## Wearable course and individualized deviations

Observations ingested: 30. Last update: 2026-01-30T06:00:00+00:00.

Flagged observations: 18. Current smoothed deviation index: 4.02.

| Synthetic sensor | First observed | Last observed |
|---|---|---|
| nremhr | 63.58 | 74.13 |
| rmssd | 32.46 | 12.10 |
| spo2 | 96.80 | 97.43 |
| full_sleep_breathing_rate | 16.27 | 17.31 |
| sleep_hours | 6.34 | 5.84 |
| stress_score | 32.18 | 41.11 |

## Model explanation

heart_disease: top reference substitutions: age (+0.016); gen_health (+0.005); bad_physical_days (+0.002); physical_activity (+0.002); bad_mental_days (+0.002)

hypertension: top reference substitutions: age (+0.045); bmi (+0.027); spo2 (-0.001); gender (+0.000); smoking_history (+0.000)

Substitution differences are non-additive associations. They are not causal effects or an individual uncertainty interval.

## Hypothetical scenarios

Hypothetical model scenario - not a validated treatment recommendation.

sleep_regularity over 14 days; assumptions: {'sleep_hours': 7.5, 'physical_activity': 1, 'rmssd': 50.0}

heart_disease: 0.023 to 0.021; change -0.002. No treatment effectiveness is inferred.

hypertension: 0.115 to 0.115; change +0.000. No treatment effectiveness is inferred.

## Evidence limitations and clinical review

Source dataset is cross-sectional with unknown device linkage, units, score direction and redistribution rights. Internal held-out classification does not establish external clinical validity. Synthetic anomalies reflect engineered perturbations.

Persistent abnormalities in real clinical practice require qualified clinical assessment. This demonstration does not prescribe medication or change treatment.